Natural Language Processing

for Social Scientists

Week 1 — Concepts, Use Cases & First Steps in R

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Four Weeks — One Arc

A progression from counting words to understanding meaning

Week 1

Concepts + First Steps

What is NLP? Method map. Use cases. The measurement problem. tidytext + quanteda intro.

TODAY

Week 2

Dictionary & Sentiment

Frequency analysis. Sentiment lexicons. Custom dictionaries. Validation.

Week 3

Topic Modelling

LDA. Structural Topic Model (stm). Covariates. Interpreting topics.

Week 4

Classification & Embeddings

Supervised text classification. Word embeddings. Introduction to language models.

What Is Text as Data?

The core shift: treating language as structured, countable, comparable material

The scale argument

A human researcher can read 10–20 documents per hour. A computer can process 10,000.

NLP makes it possible to apply social science questions — about framing, sentiment, ideology, identity — to corpora that would be impossible to read manually:

- All parliamentary speeches since 1950
- Every newspaper front page over a decade
- Millions of social media posts
- Tens of thousands of open survey responses

Types of text in social science

Political:	Speeches, manifestos, parliamentary debates, party platforms
Media:	News articles, editorials, social media, podcasts (transcribed)
Administrative:	Court decisions, policy documents, bureaucratic records
Survey:	Open-ended responses, interview transcripts, focus groups
Historical:	Archives, digitised newspapers, diaries, letters

The Method Ladder

From counting words to understanding meaning — complexity increases, so does interpretive effort

1	**Frequency & Keywords** *Word counts, tf-idf, KWIC (keyword-in-context)*	Q: What words are most distinctive of this corpus or group?	**Week 1**
2	**Dictionary & Sentiment** *Pre-built or custom lexicons applied to text*	Q: How much negativity / populism / emotion is in these speeches?	**Week 2**
3	**Topic Modelling** *LDA, Structural Topic Model — unsupervised*	Q: What themes run through this corpus, and how do they vary by time or group?	Week 3
4	**Supervised Classification** *Train a model on labelled examples, apply to new text*	Q: Does this document express support or opposition to immigration?	Week 4
5	**Embeddings & Language Models** *Word2Vec, BERT, GPT — dense semantic representations*	Q: Which documents are semantically closest to this concept?	Week 4+

Four Social Science Use Cases

Real published studies — each uses a different method family

📜 Ideological scaling of manifestos

Method: Text scaling (Wordfish)

Laver, Benoit & Garry (2003) estimated party positions from election manifestos by treating word frequencies as ideological signals. No human coding of texts required — just the words.

Reproduced results from expert surveys at a fraction of the cost and time.

🏛 Sentiment in parliamentary debates

Method: Dictionary / sentiment

Researchers tracked the emotional tone of parliamentary speeches over decades — finding that negative sentiment spikes during economic crises and correlates with populist voting.

Connects macro-political events to the language of democratic institutions.

📰 Media framing of migration

Method: Topic modelling (STM)

Heidenreich et al. (2019) used structural topic models to show that migration coverage in European newspapers frames the issue as a 'threat' significantly more often in right-wing outlets.

Scale impossible by hand: 1.3 million articles across 7 countries.

📋 Scaling up survey open-responses

Method: Supervised classification

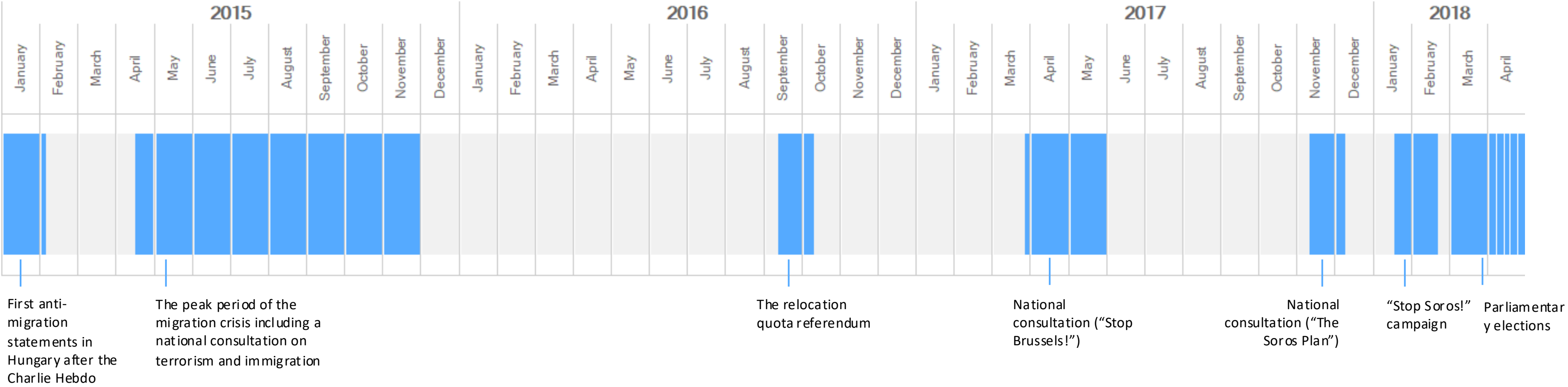
Researchers trained a classifier on a human-coded subset of open-ended responses, then applied it to the remaining 40,000 — reproducing the manual coding scheme at scale with >85% accuracy.

Bridging qualitative insight and quantitative scale.

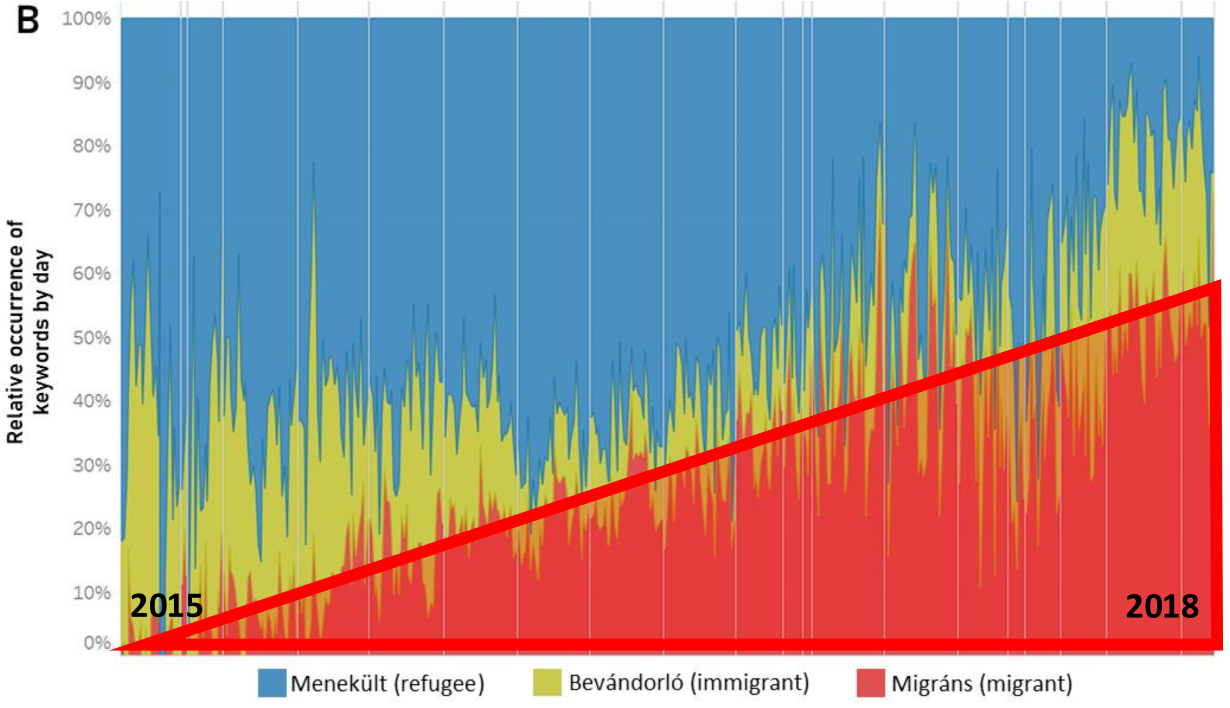
3. Methodology: Corpus

- **8 major** Hungarian online news portals (both PG and NPG)
- **31,121 news articles** covering period of 07 January 2015 to 14 April 2018 in seven non-consecutive sections from the aftermath of most major international or domestic events related to migration
- 62% of news items from PG sources ⇔ 38% of news items from NPG sources

- Total number of words: **approx. 15 million**
  - 5,733 compound words (tokens)
  - 193 metaphorical compound types
- Ratio of metaphorical compounds: ***migráns*** (25.8%), *menekült* (14.5%), *bevándorló* (6.4%)



2. The Hungarian context
Case of online media



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Relative occurrence of keywords by period

Keyword	Period							Total
	1	2	3	4	5	6	7	
Menekült (refugee)	59%	64%	51%	43%	46%	42%	20%	**58%**
Bevándorló (immigrant)	36%	16%	14%	15%	18%	21%	28%	**17%**
Migráns (migrant)	5%	21%	35%	42%	36%	37%	53%	**26%**
Total	100%	100%	100%	100%	100%	100%	100%	100%

4. Results

Table 2. Occurrence of identified source domains in compounds with *bevándorló / menekült / migráns* as modifiers (in % and log odds ratios)

	Source domains appearing in the compounds	*bevándorló* ("immigrant") %	*bevándorló* ("immigrant") wlo	*migráns* ("migrant") %	*migráns* ("migrant") wlo	*menekült* ("refugee") %	*menekült* ("refugee") wlo	Total %
1	Flood	65.4	+4.58	58.2	−8.25	74.5	+3.80	70.7
2	Object	15.4	+0.75	15.2	−2.54	23.1	+1.49	21.2
3	Business	0	–	8.4	+8.23	0.8	−0.69	2.5
4	Crime	8.7	+0.779	8.5	+5.20	0.3	−5.15	2.3
5	War	5.8	+0.236	6.2	+3.72	0.7	−3.39	2.0
6	Animal	2.9	+0.944	1.3	+1.69	0.1	−2.13	0.5
7	Pressure	0	–	0.7	+1.97	0.3	+0.78	0.4
8	Building	0	–	0.8	+2.37	0.1	+0.01	0.3
9	Misc (*monster /natural force/ horror/game)*	1.9	+0.527	0.6	+0.71	0.1	−0.99	0.2
10	Disease	0	–	0.2	+2.04	0	*NA*	0.1
	Total (%)	100%		100%		100%		100%
		n = 104		*n* = 1,276		*n* = 4,353		*n* = 5,733

Weighted log odds ratios (wlo)

The higher the log odds ratio, the more specific the keyword is to a source domain, as compared to the others. A negative log odds ratio means that a keyword is less likely to appear with a given source domain.

4. Results

Keyword	Source domain		Period 1-2 (2015)	Period 3 (Oct-Nov 2016)	Period 4 (Mar-Apr 2017)	Period 5 (Nov-Dec 2017)	Period 6 (Jan-Feb 2018)	Period 7 (Mar-Apr)
menekült	flood		78.3%	56.6%	47.9%	40.3%	52.4%	45.7%
	object		20.0%	41.7%	45.3%	58.1%	42.9%	52.9%
	business		0.6%	0.4%	6.0%	1.6%	4.8%	1.4%
	war		0.7%	1.3%	0.9%	0%	0%	0%
	crime		0.3%	0%	0%	0%	0%	0%
	Total	(%)	100%	100%	100%	100%	100%	100%
		(n)	3,786	228	117	62	63	70
migráns	flood		89.4%	66.7%	29.5%	37.5%	17.1%	20.5%
	object		6.7%	19.3%	9.8%	50.0%	48.8%	22.1%
	business		1.1%	0%	54.9%	0%	3.7%	1.6%
	war		2.5%	10.5%	4.0%	8.3%	3.7%	17.4%
	crime		0.4%	3.5%	1.7%	4.2%	26.8%	38.4%
	Total	(%)	100%	100%	100%	100%	100%	100%
		(n)	566	171	173	48	82	190

„Refugee deluge”
„Refugee wave”

„Refugee quota”

„Migrant invasion”
„Migrant attack”

„Migrant deluge”
„Migrant wave”

„Migrant quota”

„Migrant business”

„Migrant gang”
„Migrant violence”
„Migrant smuggler”
„Migrant criminal”

4. Results

Table 4. Use of identified source domains by keywords in PG and NPG media sources

Media source		Keyword: Bevándorló *(immigrant)*	Menekült *(refugee)*	Migráns *(migrant)*	
PG	%	2.6%	61.8%	35.6%	100%
	wlo	+15.23	−53.75	+34.17	
NPG	%	1.2%	87.0%	11.8%	100%
	wlo	−13.61	+51.58	−30.17	

- Significant association between the use of keywords with identified source domains in PG and NPG media sources
- PG: *migráns* ⇔ NPG: *menekült*

Table 5. Use of identified source domains by keywords in PG an

Source domain	Media source: Pro-government % *(n)*	*wlo*	Non pro-government % *(n)*	*wlo*
flood	66.1% (1466)	−3.73	73.6% (2585)	3.49
object	21.6% (479)	0.24	20.9% (735)	−0.22
business	3.8% (85)	1.75	1.7% (58)	−1.65
war	2.4% (54)	1.21	1.2% (42)	−1.13
crime	4.9% (108)	3.47	0.6% (22)	−3.43
animal	0.5% (10)	−0.86	1.1% (37)	0.75
pressure	0.4% (9)	0.13	0.3% (12)	−0.12
building	0.1% (3)	−0.52	0.3% (12)	0.45
misc	0.1% (2)	−0.55	0.3% (10)	0.47

The Measurement Problem

The most important slide in this course

NLP output

— Sentiment score

— Topic proportion

— Word frequency

— Classifier label

$\neq$

Your construct

— Attitude

— Social phenomenon

— Meaning

— Behaviour

Every NLP measure needs a validity argument. Ask: does this operationalisation actually capture the concept I care about — in this language, this domain, this time period?

Limits and Honest Expectations

NLP is a powerful tool — not a shortcut. Each limitation has a practical implication.

🌍 Language & domain dependency

Models trained on English Twitter perform poorly on Hungarian parliamentary text. Pre-trained models embed the cultural assumptions of their training data. Always validate on your specific corpus.

🧹 Cleaning takes longer than analysis

OCR errors, encoding issues, inconsistent formatting, domain-specific stopwords — text data is rarely clean. Expect 60–70% of project time to be preprocessing, not modelling.

📦 Black boxes and reproducibility

Large language models are difficult to interpret. Results can change with model versions. Always report model, version, hyperparameters, and preprocessing decisions. Reproducibility is a real challenge.

📏 Quantity ≠ quality

A million documents does not guarantee valid inference. Biased corpora, missing data (who does not speak?), and survivorship bias are all real threats to external validity.

🎯 NLP is a tool, not a theory

The research question, the theoretical framework, and the interpretation must still come from you. NLP operationalises concepts — it does not generate them.

Discussion

5 minutes individual thinking · 10 minutes group discussion

1

What text data exists — or could exist — in your own research area?

Think about what you would want to measure from it. Is the text already collected, or would you need to create/gather a corpus? What would be the main obstacles?

2

Pick a concept from your field. How would you operationalise it from text — and what could go wrong?

Examples: trust, polarisation, framing, social cohesion, inequality. What NLP method would you use? What would a failure of validity look like in practice?

Coding Block 1 — The tidytext Data Structure

One token per row — text as a tidy tibble

Raw format

```
doc_id  year  text
1       1801  'Fellow citizens, called...'
2       1802  'When we assemble together...'
3       1803  'In calling you together...'
```

→ Each document is one row.
Text is one long string.
Hard to analyse directly.

unnest_tokens()

Tidy text format (one token per row)

```
doc_id  year  word
1       1801  fellow
1       1801  citizens
1       1801  called
2       1802  when
2       1802  we
2       1802  assemble
```

→ Every row is one word.
Works with dplyr, ggplot2.
Count, join, filter naturally.

Coding Block 2 — The SOTU Pipeline

Load → tokenise → remove stopwords → count → tf-idf → visualise

```
library(tidyverse)
library(tidytext)
library(sotu)

# Load SOTU corpus (all addresses 1790-2020)
sotu_df <- sotu_meta |>
  bind_cols(tibble(text = sotu_text)) |>
  mutate(era = case_when(
    year < 1850 ~ 'Early Republic (1790-1849)',
    year < 1900 ~ 'Gilded Age (1850-1899)',
    year < 1945 ~ 'Progressive & War (1900-1944)',
    year < 1990 ~ 'Cold War (1945-1989)',
    TRUE        ~ 'Contemporary (1990-2020)'
  ))

# Tokenise and remove stopwords
tidy_sotu <- sotu_df |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = 'word')

# Compute tf-idf by era
sotu_tfidf <- tidy_sotu |>
  count(era, word, sort = TRUE) |>
  bind_tf_idf(word, era, n) |>
  arrange(desc(tf_idf))
```

sotu package

One-line corpus load — all presidential addresses as plain text

Era grouping

Five temporal periods — our unit of comparison for tf-idf

unnest_tokens()

Splits each speech into one-word-per-row

anti_join(stop_words)

Removes 'the', 'and', 'of' — words with no discriminating power

bind_tf_idf()

Computes term frequency × inverse document frequency per era

Coding Block 3 — tf-idf Results & Visualisation

Most distinctive words per presidential era — what this reveals sociologically

```
# Visualise top terms per era
sotu_tfidf |>
  group_by(era) |>
  slice_max(tf_idf, n = 8) |>
  ungroup() |>
  mutate(word = reorder_within(word,
                  tf_idf, era)) |>
  ggplot(aes(x = word, y = tf_idf,
             fill = era)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~era, scales = 'free') +
  coord_flip() +
  scale_x_reordered() +
  theme_minimal(base_size = 10) +
  labs(title = 'Most distinctive words by era',
       x = NULL, y = 'tf-idf score')
```

Discussion: what does the shift in vocabulary tell you about the changing role of the US state?

What you will see — and why it matters

Early Republic:

'commerce', 'Indians', 'militia', 'revenue'

Institutional vocabulary of a new state

Gilded Age:

'tariff', 'silver', 'labor', 'railroad'

Industrial economy, class conflict

Progressive/War:

'naval', 'allies', 'armistice', 'soviet'

World wars dominate the lexicon

Cold War:

'soviet', 'nuclear', 'communist', 'vietnam'

Ideological competition as the frame

Contemporary:

'terrorism', 'recession', 'healthcare', 'climate'

Crises define contemporary discourse

Coding Block 4 — Introduction to quanteda

A second ecosystem — corpus → tokens → DFM (document-feature matrix)

```r
library(quanteda)
library(quanteda.textstats)
library(quanteda.textplots)

# Build a quanteda corpus from the SOTU data
sotu_corpus <- corpus(sotu_df,
  docid_field = 'president',
  text_field  = 'text')

# Tokenise and clean
sotu_tokens <- sotu_corpus |>
  tokens(remove_punct = TRUE,
         remove_numbers = TRUE) |>
  tokens_remove(stopwords('en'))

# Build Document-Feature Matrix (DFM)
sotu_dfm <- dfm(sotu_tokens)

# Top features across corpus
topfeatures(sotu_dfm, 15)

# Word cloud by era
dfm_group(sotu_dfm,
          groups = sotu_df$era) |>
  textplot_wordcloud(comparison = TRUE)
```

The DFM: documents × features

	war	trade	economy	freedom
Adams 1800	0	3	1	2
Lincoln 1863	5	0	2	4
FDR 1942	12	1	3	1
Obama 2010	2	2	5	3

Each row = one document
Each column = one word (feature)
Each cell = count (or tf-idf weight)

This matrix is the input to almost all advanced NLP methods:
LDA, SVD, classifiers, scaling.

tidytext → tidy tibble (one token per row) · quanteda → DFM (document × feature matrix)

tidytext vs. quanteda — When to Use Which

Both packages are excellent. They solve different problems and work best together.

	tidytext	quanteda
Data structure	Tidy tibble (one-token-per-row)	DFM (sparse matrix)
Integrates with	dplyr, ggplot2, tidymodels	quanteda.textmodels, stm, spacyr
Strengths	Exploration, visualisation, pipeline clarity	Speed on large corpora, rich NLP tools
Best for	Teaching, EDA, joining with metadata	Topic models, scaling, classification
Corpus sizes	Small–medium (millions of tokens)	Large corpora (billions of tokens)
Learning curve	Gentle — uses tidyverse idioms	Moderate — new data structures
This course	**Weeks 1–2 (primary)**	**Weeks 3–4 (primary)**

What Comes Next

The method ladder — where today sits, and where we are going

Week 1

Concepts + First Steps

— Text as data · the method ladder · use cases

— Measurement validity · limits

— tidytext pipeline · tf-idf · quanteda intro

DONE TODAY ✓

Week 2

Dictionary & Sentiment

— AFINN · Bing · NRC lexicons

— Custom dictionaries

— Validation: does the lexicon travel?

Week 3

Topic Modelling

— LDA: what it does

— Structural Topic Model (stm)

— Topic prevalence as outcome variable

Week 4

Classification & Beyond

— Supervised classification with tidymodels

— Word embeddings: word2vec

— Introduction to transformer models

Take-Home Exercise

Complete before Week 2 · bring your output and interpretation to class

1

Change the grouping variable

Re-run the tf-idf analysis grouping by individual president rather than era. Which presidents have the most distinctive vocabularies? Does this align with your prior knowledge of US political history?

2

Narrow the time window

Filter the corpus to just the last 30 years (1990–2020). Re-run tf-idf by era — do the results change meaningfully? What does this tell you about within-era variation?

3

Write your interpretation

Write 3–4 sentences: What do the most distinctive words reveal about how political discourse has changed? What cannot you conclude from this analysis — and why? What would you need to claim causality?